SHAHIL SHAIK

Ph.D. Researcher in Robotics and Intelligent Systems

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RESEARCH PROFILE

Robotics Ph.D. researcher developing efficient learning architectures for embodied and multi-agent intelligence. Research spans transformer attention, vision-language and vision-language-action models, multi-agent deep reinforcement learning, distributional policy optimization, and shared human–robot control. Published in the American Control Conference and IET Cyber-Systems and Robotics; experienced in reproducible, GPU-accelerated research with PyTorch.

EDUCATION

2021–Present — **Ph.D. in Mechanical Engineering (Robotics)**
Clemson University, Clemson, South Carolina
Research focus: intelligent multi-agent robotic systems and learning-based autonomy.

2019–2021 — **Graduate Study in Mechanical Engineering**
Clemson University, Clemson, South Carolina
Entered the M.S. program in 2019 and transitioned to the Ph.D. program in 2021.

2015–2019 — **Bachelor's Degree in Mechanical Engineering**
Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam, India
GPA: 8.55/10.

RESEARCH INTERESTS

Efficient transformer architectures • Vision-language-action learning • Multi-agent reinforcement learning • Robot learning • Human–robot shared control • Distributed and memory-efficient deep learning

SELECTED RESEARCH

Lie-Generated Metric Attention (LGMA) | *Lead developer and investigator*
Developing a PyTorch attention architecture that generates head-specific metrics from shared Lie-algebra generators to reduce parameter and key–value cache costs. Built controlled MHA-versus-LGMA experiments, stability diagnostics, checkpointing, efficiency accounting, and single-/multi-GPU training workflows for TinyStories and robot-learning studies.

Vision-Language(-Action) Models for Multi-Agent Robotics | *Research and software development*
Investigating multimodal critics and action models for policy evaluation and robot learning, including MA-VLCM and planned LIBERO benchmarking. Work integrates visual observations, language-conditioned reasoning, temporal context, and multi-agent state representations.

Multi-Agent Deep Reinforcement Learning | *Doctoral research*
Developed methods for learning under constrained inter-agent communication and studied distributional policy-gradient estimation. Research emphasizes coordination, robustness, sample efficiency, and principled evaluation across multi-agent settings.

Shared Control of Mobile Robots | *Collaborative research*
Studied deep reinforcement-learning approaches that combine human guidance with autonomous control for mobile robot navigation, resulting in a peer-reviewed journal publication.

PUBLICATIONS AND PREPRINTS

Shaik, S., Smereka, J. M., & Wang, Y. (2026). *Multi-Agent Deep Reinforcement Learning Under Constrained Communications*. arXiv preprint arXiv:2601.17069.

Shaik, S., Parameshwaran, A., Nayak, A., Smereka, J. M., & Wang, Y. (2026). *MA-VLCM: A Vision Language Critic Model for Value Estimation of Policies in Multi-Agent Team Settings*. arXiv preprint arXiv:2603.15418.

Nayak, A., **Shaik, S.**, & Wang, Y. (2026). *Belief-Aware VLM Model for Human-like Reasoning*. arXiv preprint arXiv:2604.09686.

Shaik, S., Smereka, J. M., & Wang, Y. (2025). *Generalized Advantage Estimation for Distributional Policy Gradients*. Proceedings of the American Control Conference, 3073–3078.

Tian, C., **Shaik, S.**, & Wang, Y. (2021). *Deep Reinforcement Learning for Shared Control of Mobile Robots*. IET Cyber-Systems and Robotics, 3(4), 315–330.

RESEARCH SOFTWARE AND REPRODUCIBILITY

Open-source LGMA implementation. Maintains a public PyTorch research prototype with interchangeable LGMA and conventional multi-head attention, synthetic and TinyStories experiments, and VLA data/model/evaluation modules. Repository

Experiment integrity. Built unit and smoke tests, configuration-driven experiments, resumable checkpoints, validation workflows, per-GPU throughput and memory accounting, and automated summaries for validation-loss-versus-token and validation-loss-versus-GPU-hour comparisons.

Research-computing readiness. Prepared bf16 CUDA training, gradient accumulation, single- and multi-GPU DistributedDataParallel execution, Slurm launch patterns, dataset staging, and portable Conda/Apptainer environment plans for accelerator-based studies.

TECHNICAL SKILLS

Programming	Python; scientific computing and research software development
Deep Learning	PyTorch; transformer models; attention mechanisms; reinforcement learning; VLM/VLA
GPU & Scale	CUDA; bf16 mixed precision; PyTorch DistributedDataParallel; torchrun; Slurm
Research Stack	Hugging Face Datasets/Transformers; NumPy; h5py; OpenCV; Weights & Biases
Robotics & Tools	MuJoCo/LIBERO workflows; Git/GitHub; Conda; Apptainer; pytest